

Daily Progress & Next Steps Plan

Constraint-Interaction Metric for Symbolic Regression

Biswajyoti Nath

Subinoy Nath

July 16, 2026

Part 1: Summary of Today's Work (Pipeline Audit & Hardening)

Today's focus was executing a rigorous architectural audit of the `src/analysis.py` and `src/experiments.py` pipelines in response to the Go/No-Go pre-matrix review. We successfully hardened the infrastructure to resolve all six blocking and non-blocking issues, bringing the project to a definitive "Go" state for the full empirical benchmark suite.

- **JIT Warmup Handling:** Implemented a robust exclusion layer in `compute_S_ij` to automatically drop `jit_warmup_suspect` rows, ensuring Julia compilation overhead does not contaminate the wall-clock speedup metrics.
- **Constraint Enforcement Gap (C1b):** Identified a discrepancy between the theoretical SymPy-based consecutive binary constraint (C1b) and PySR's actual search limitations. We officially split C1 into C1a (nested trig) and C1b, and disabled C1b via the `enforce_c1a_only` config flag, perfectly aligning the $M(i, j)$ theoretical density with the actual search distribution PySR evaluates.
- **Distribution Audit:** Wrote `scripts/distribution_audit.py` as a 1-iteration PySR proxy to capture early-generation expressions. The audit confirmed empirical constraint densities diverge from Monte Carlo uniform estimates by less than an order of magnitude (maximum $2.94\times$), formally establishing that $M(i, j)$ retains directional predictive power.
- **Statistical Power & Equivalence Criteria:** Updated the PRD with attrition-sensitive power calculations, confirming > 0.80 power up to a 30% timeout rate. We also formally demoted symbolic equivalence (`check_semantic_exact`) to an exploratory role, locking in numerical relative MSE as the sole primary recovery criterion.
- **Testing & Stability:** Added verification tests for JIT exclusion and verified that the entire test suite (60 tests spanning generation, constraints, experiments, and analysis) is 100% passing and completely deterministic.

Part 2: Strict Plan for Tomorrow (Matrix Execution & Handbook Drafting)

With all known methodological biases and infrastructural bugs systematically resolved, the pipeline is fully prepared to execute the $4 \times 12 \times 5 = 240$ scenario matrix.

- **Phase 1 – Execution:** Execute the full experimental matrix across all datasets and constraints using `src/experiments.py`. Monitor progress and ensure `metadata.json` accurately logs git commit hashes and configurations.

- **Phase 2 – Analysis:** Run `src/analysis.py` over the completed `pysr_results.csv` to calculate Pearson r and p -values between the theoretical metric $M(i, j)$ and the empirical search cost $S(i, j)$.
- **Phase 3 – Reporting:** Generate all required visualisations (scatter plots, interaction heatmaps) and begin compiling the final 55-65 page implementation handbook for submission, integrating our detailed LOGBOOK findings.